The Impact of Sycophantic AI on Attitudes and Decisions

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Abstract: There is widespread concern that AI chatbots are "sycophantic," or overly agreeable and validating. However, less is known about the causal impact of interacting with sycophantic AI on beliefs and behaviors. Across seven studies (total n = 7,227), we found that people enjoyed interacting with sycophantic AI chatbots more than interacting with neutral chatbots or "disagreeable" chatbots that challenged them. Brief conversations with sycophantic chatbots about political or personal topics increased the strength of people's attitudes. However, popular AI models not prompted to be sycophantic did not consistently change attitudes. Sycophantic chatbots also inflated people's perceptions that they were better than the average person on desirable traits (e.g., intelligence, empathy), and also led people to bet more money that they scored better than average on tasks purporting to measure desirable traits. Participants consistently rated sycophantic chatbots as more "unbiased" than disagreeable chatbots, even though third-party raters viewed these chatbots as equally biased, suggesting that people may be blind to biases in AI output that aligns with their views. Altogether, these results suggest that sycophantic model behavior can entrench attitudes and lead to overconfident beliefs and decisions. Despite this, people enjoy sycophantic models and often fail to recognize their biases.

Main Text:

There has been growing concern that artificial intelligence (AI) chatbots are "sycophantic," or excessively agreeable and validating[1,2]. Public concern about AI sycophancy intensified following accusations that OpenAI's popular model ChatGPT-4o had become overly sycophantic in April 2025, prompting the company to roll back changes to this model and implement new safeguards[3]. Recent research found that large language models (LLMs) are roughly 50% more sycophantic than humans[4], and can affirm harmful or delusional ideas[5,6].

Despite research documenting sycophancy in AI models[7], little work has systematically examined the psychological and behavioral consequences of interacting with sycophantic AI across a wide range of topics. Given the rapid growth of AI tools for personal and professional decision-making, it is crucial to understand the impact of AI sycophancy. Here, we investigated whether interactions with sycophantic AI entrenched attitudes and led to overconfident beliefs and decisions. Moreover, we examined whether users enjoyed interacting with sycophantic AI, whether users could recognize its biases, and whether it was possible to build non-sycophantic AI that users still enjoyed.

A growing body of research has found that AI can persuade people to hold more accurate viewpoints or consider opposing perspectives. For instance, dialogues with chatbots designed to "debunk" conspiracies can decrease people's belief in conspiracy theories[8]. Chatbots can also successfully persuade people about political issues[9,10]. Some studies suggest that LLMs may rival or outperform incentivized human participants and experts in crafting persuasive messages[11–13].

Despite AI's capacity to persuade people to hold different or more accurate beliefs, there is extensive evidence that people often prefer to have their pre-existing beliefs validated[14–17]. Decades of research on confirmation bias[18] and selective exposure[14,19] suggests that people seek out and prefer information that confirms their pre-existing beliefs and aligns with their identities[20,21]. As a result, people might prefer to interact with AI models that validate rather than challenge their viewpoints. Furthermore, exposure to belief-confirming information from AI models may increase attitude polarization[14,19,22,23].

Sycophantic AI may also lead to overconfident beliefs and behaviors. People tend to think they are better than the average person on a number of desirable traits, such as intelligence and empathy: a phenomenon called the "better-than-average" effect[24]. Since it is a statistical impossibility for more than half of people to be better than average, this finding suggests that people are overly positive or overconfident in their self-perceptions[25]. Sycophantic chatbots may amplify the "better-than-average" effect by flattering people's ideas and personal qualities.

Sycophancy has been defined in a variety of different ways, and many consider it to be a multi-dimensional and fragmented construct[2,26–28]. We tested the impact of two components of sycophancy: social validation and the provision of one-sided information. While some papers define sycophancy as a model agreeing with a verifiably incorrect response (sometimes called "factual sycophancy"),[28,29] we analyzed sycophancy regarding political and social topics that did not have a ground truth. Recent research has suggested that AI is primarily persuasive because of its ability to provide targeted facts and evidence[30]. Thus, the selective presentation of factual statements may explain why sycophantic chatbots entrench attitudes. However, given past research on the impact of flattery[31], compliments[32], and ingratiation[33], we expected that people would enjoy validating chatbots more regardless of whether they provided factual information.

Overview. Across seven studies (n = 7,227), we investigated the psychological and behavioral consequences of AI sycophancy. In **Studies 1-3**, we tested the impact of interacting with sycophantic and non-sycophantic (or "disagreeable") AI chatbots while discussing polarized political issues on attitude extremity, attitude certainty, enjoyment, and perceptions of AI bias. In **Study 3**, we disentangled two key components of sycophancy by testing the effects of sycophantic and disagreeable chatbots that either did or did not provide factual information. In **Studies 4-5**, we had annotators, fact-checkers, and LLMs rate the conversations along various dimensions to better understand their content. In **Study 6**, we tested whether the effects generalized to non-political topics, persisted over time, affected behavior, and were robust to a new control condition. Finally, in **Study 7**, we tested whether it was possible to make chatbots that present opposing information that people are more receptive to.

Sycophantic AI and Political Topics

In Studies 1-3 (n = 3,283), we tested the causal impact of sycophantic, disagreeable (or non-sycophantic), and undirected chatbots (e.g., chatbots with the default settings) on political attitudes. In each study, participants engaged in a three-turn dialogue with an AI chatbot about various political topics (gun control in Studies 1 and 3; abortion, immigration, or healthcare in Study 2). The AI chatbots were powered by commercial large language models (ChatGPT-4o in Study 1; ChatGPT-5 in Study 2; and Claude Sonnet 4; Gemini 2.5 Flash; or ChatGPT-5 in Study 3) that were prompted to behave in a specific way.

Experimental Conditions. US participants were randomly assigned to interact with 1) a sycophantic chatbot that was prompted to affirm and validate the participant's perspective, 2) a disagreeable chatbot that was prompted to challenge the participant's perspective, or 3) an undirected chatbot that was not instructed to behave in any particular way, but was sometimes given some minimal context about the interaction or topic discussed (see ***Supplementary Appendix*** for table of prompts used across studies). In an additional control condition designed

to test the comparative impact of discussing an unrelated topic, participants discussed the benefits of owning dogs and cats with an undirected AI chatbot, following prior work[8].

Validating the Manipulations. An additional study (Study 4; see ***Supplementary Appendix***) with human annotators validated that these conditions successfully manipulated sycophancy, or the amount of social and emotional validation in text (see ***Extended Data Fig 1***). To test whether real-world chatbot conversations can be as sycophantic as the sycophantic chatbots in our experiments, we analyzed a separate dataset[34] of 9,837 naturalistic ChatGPT interactions using LLM prompts that were highly correlated with judgments of human annotators. Broadly, we found that the chatbots in the sycophantic condition were, on average, more sycophantic than naturalistic LLM replies across a variety of dimensions (see ***Supplementary Appendix*** for details). However, there was also considerable overlap: naturalistic ChatGPT replies were approximately as agreeable as the sycophantic chatbot in 72% of turns, and were approximately as validating in 8% of turns. In conversations that were at least 25 turns long, 46% contained at least one chatbot response that was approximately as validating as the typical response in the sycophantic condition. Thus, while the sycophantic condition tested the effects of "strong" sycophancy (as opposed to default LLM behavior), our analysis, combined with analyses from several other papers[5,35,36], suggests that chatbots can display strong sycophancy frequently enough to warrant concern about the causal effects of strong sycophancy.

Outcomes. Participants rated their attitudes and attitude certainty on a 0-100-point scale (0 = strongly oppose/very uncertain; 100 = strongly support/very certain) before and after each conversation. Afterwards, they completed additional measures assessing perceptions of the AI chatbot (enjoyment and perceptions of bias) and self-perceptions on a 1-6 Likert scale (1 = strongly Disagree; 6 = strongly agree). Because results were relatively consistent across studies, we present the pooled results of Studies 1-3 in the main text. See the ***Supplementary Appendix*** for all prompts used, all outcome variables, results for individual studies, and robustness checks. All individual studies were pre-registered, but the pooled analysis was not.

Attitude Change. As predicted, the sycophantic chatbot increased the extremity of people's beliefs (e.g., the distance from the midpoint on a 100-point scale) as compared to the control condition (b = 2.03, 95% CI [1.27, 2.79], $p < .001$). By contrast, the disagreeable chatbot decreased the extremity of people's beliefs (b = -1.58, 95% CI [-2.57, -0.58], p = .002). Similarly, the sycophantic chatbot increased attitude certainty by about four percentage points relative to the control condition (b = 3.78, 95% CI [2.77, 4.79], $p < .001$) whereas the disagreeable chatbot decreased it by about 1.5 percentage points (b = -1.46, 95% CI [-2.73, -0.18], p = .025). In other words, sycophantic AI made people's attitudes more extreme, and disagreeable AI made them less extreme. The undirected AI did not change attitude extremity as compared to the control condition (p = .290), although it did slightly increase attitude certainty (b = 2.02, 95% CI [0.70, 3.34], p = .003).

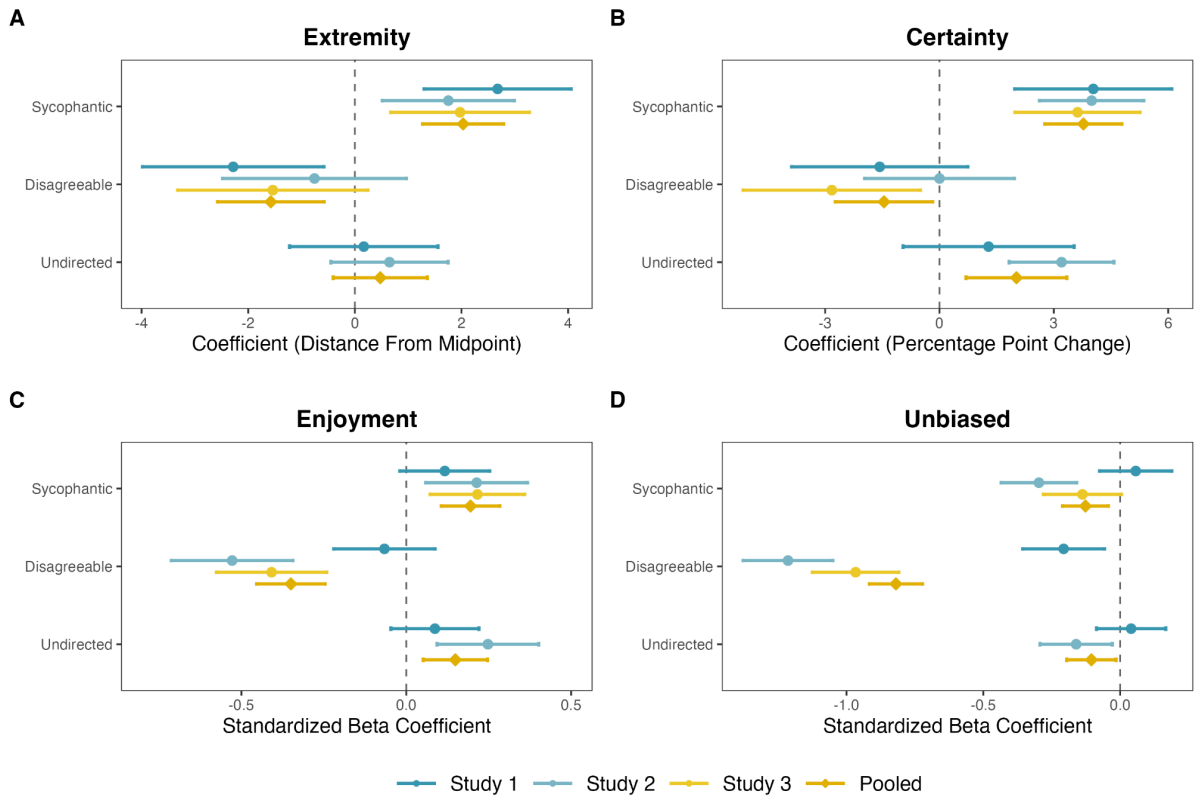


Fig. 1. Effects of chatbots on main outcome variables for Studies 1-3. *As shown, sycophantic chatbots increased extremity (Panel A) and certainty (Panel B), whereas disagreeable chatbots decreased extremity and certainty. Sycophantic chatbots were enjoyed more (Panel C) and viewed as less biased (Panel D) than disagreeable chatbots. The control condition serves as the reference group in all models, and for Study 3, the "non-fact-providing" conditions are not shown, though those are shown separately in* ***Fig. 2****, and separate estimates with the non-fact-providing chatbots are also shown in the* ***Supplementary Appendix****. Attitude extremity coefficients reflect distance from the midpoint (50) on a 0-100 scale, and attitude certainty coefficients reflect absolute changes on a 0-100 scale. Attitude extremity and certainty coefficients are unstandardized. Enjoyment and bias were rated on a 1-6 Likert scale, and standardized coefficients are shown for these variables. Error bars represent 95% confidence intervals.*

Enjoyment. Participants enjoyed the sycophantic chatbot more than the chatbot in the control condition ($\beta = 0.19$, 95% CI [0.11, 0.28], $p < .001$) and the disagreeable chatbot ($\beta = 0.54$, 95% CI [0.44, 0.65], $p < .001$). However, there was no difference in enjoyment between the sycophantic and undirected conditions ($\beta = 0.05$, 95% CI [-0.05, 0.14], $p = .352$).

In Study 2, we provided people with the option to choose to have another conversation with the same chatbot again or simply end the survey earlier — a behavioral outcome mirroring

real-world decisions to continue or end chatbot conversations. We found that people were nine percentage points more likely to choose to interact with the sycophantic chatbot again compared to the disagreeable chatbot (b = 0.09, 95% CI [0.02, 0.16], p = .017). While the sycophantic and undirected chatbots did not differ from the control condition (ps > .360), people were less likely to choose to continue the conversation with the disagreeable chatbot than the control condition (β = -0.13, 95% CI [-0.20, -0.05], p < .001).

Perception of Bias. The sycophantic (β = -0.13, 95% CI [-0.21, -0.04], p = .003) and disagreeable (β = -0.82, 95% CI [-0.92, -0.72], p < .001) chatbots were both viewed as less objective than the chatbots in the control condition. The sycophantic chatbot was not viewed as significantly different from the undirected chatbot in perceptions of bias (p = .652). The sycophantic chatbot was viewed as more objective than the disagreeable chatbot (β = 0.69, 95% CI [0.59, 0.79], p < 0.001). In other words, people believed that chatbots prompted to be sycophantic were approximately as objective as the undirected chatbots, and considerably more objective than the disagreeable chatbots.

Third-Party Perceptions of Bias. While people interacting with sycophantic chatbots considered sycophantic replies to be unbiased, an additional study (Study 4; n = 403) found that an independent sample of third-party annotators had very different perceptions. Specifically, when third-party annotators rated how unbiased (or objective) replies were across conditions in a within-subjects design, no significant differences emerged between the sycophantic and disagreeable conditions (p = .293). Moreover, third-party annotators rated the sycophantic chatbot as less objective than the undirected chatbot (β = -0.97, 95% CI [-1.23, -0.71], p < .001), even though these conditions did not statistically differ for those interacting with sycophantic chatbots in Studies 1-3. In other words, while users who interacted with sycophantic chatbots viewed them as more objective and unbiased than disagreeable chatbots, third-party annotators recognized that sycophantic and disagreeable chatbots were equally biased. This demonstrates a potential "bias blind spot" in perceptions of AI sycophancy whereby users are better at recognizing when AI is biased in favor of others as opposed to themselves.

The Better-Than-Average Effect. We also assessed the extent to which participants felt "better than average" on six positive traits (i.e., intelligence, empathy, morality, etc.). We averaged the six items to create a composite better-than-average index. The sycophantic AI led people to rate themselves higher on these traits compared to the control condition (β = 0.12, 95% CI [0.03, 0.21], p = .011), whereas the disagreeable AI led people to rate themselves lower on these traits as compared to the control condition (β = -0.12, 95% CI [-0.23, -0.02], p = .016). While these effects were small, they were consistent: the contrast between the sycophantic and disagreeable conditions was significant for each of these six traits (see ***Supplementary Appendix***). These results suggest that sycophantic AI leads to inflated self-perceptions.

Dissociating Key Dimensions of Sycophancy. Because sycophancy is a multi-dimensional construct, we disentangled the effects of two different components of sycophancy in Study 3 (n = 1,183): validation versus the one-sided presentation of factual statements. To do this, we tested the impact of sycophantic and disagreeable chatbots that were prompted to either provide factual information or not provide factual information (see prompts in the ***Supplementary Appendix***). A validation study with human annotators (Study 4) confirmed that the experimental manipulations successfully manipulated the provision of factual statements. An additional study with fact-checkers (Study 5, $n = 20$) confirmed that most facts presented by the chatbots across conditions were largely accurate ($M = 8.43$, $SD = 1.53$), with no significant differences across conditions.

As predicted, the sycophantic fact-providing chatbot increased attitude extremity ($b = 3.51$, 95% CI [1.65, 5.38], $p < .001$) and certainty ($b = 6.45$, 95% CI [4.05, 8.84], $p < .001$) as compared to the disagreeable fact-providing chatbot (see ***Fig. 2***). However, when the sycophantic and disagreeable chatbots provided no facts, we did not detect a significant difference between conditions on extremity and certainty ($p = 0.263$). When comparing the two sycophantic chatbots, the sycophantic fact-providing chatbot led to increased certainty as compared to the sycophantic non-fact-providing chatbot ($b = 2.42$, 95% CI [0.75, 4.08], $p = .005$) though this contrast was not statistically significant for attitude extremity ($b = 1.27$, 95% CI [-0.12, 2.66], $p = .074$). When comparing the two disagreeable chatbots, the disagreeable fact-providing chatbot led to decreased extremity ($b = -1.99$, 95% CI [-3.92, -0.07], $p = .042$) and certainty ($b = -2.93$, 95% CI [-5.49, -0.38], $p = .025$) compared to the non-fact-providing disagreeable chatbot. While not all contrasts are significant, this overall pattern of results provides some evidence that the sycophantic and disagreeable chatbots lead to more attitude change when accompanied by factual information.

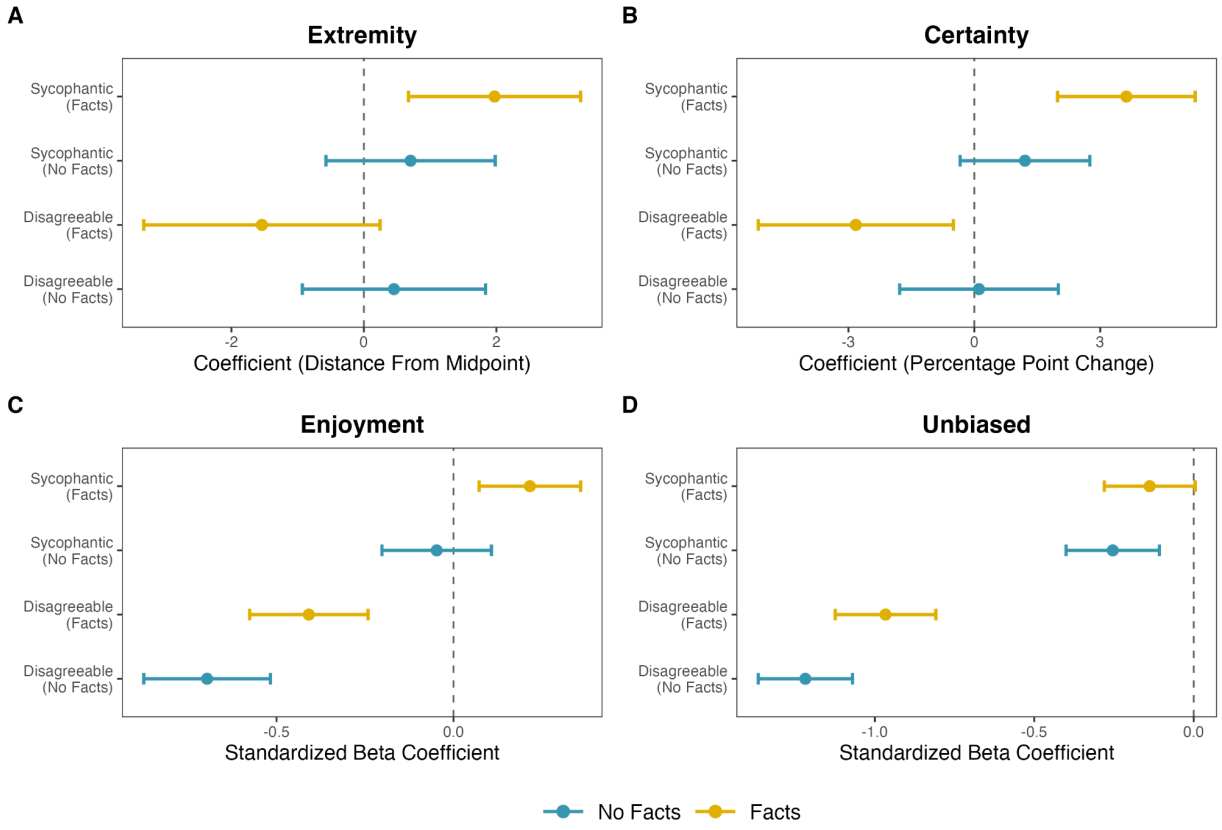


Fig. 2. Different types of sycophancy had different effects. *When sycophantic and disagreeable chatbots were prompted to also provide factual statements, they led to changes in attitude extremity (Panel A) and certainty (Panel B). When prompted to provide no factual statements, sycophantic and disagreeable chatbots did not change extremity or certainty. However, regardless of whether they provided facts, sycophantic chatbots were enjoyed more (Panel C) and were viewed as less biased (Panel D) than disagreeable chatbots. Error bars represent 95% confidence intervals.*

Participants enjoyed the sycophantic chatbots more than the disagreeable chatbots both when they provided facts ($\beta = 0.63$, 95% CI [0.46, 0.80], $p < .001$) and when they provided no facts ($\beta = 0.65$, 95% CI [0.46, 0.83], $p < .001$). The sycophantic chatbots were also viewed as less biased than the disagreeable chatbots both when the chatbots provided facts ($\beta = 0.84$, 95% CI [0.67, 1.00], $p < .001$) and when they did not ($\beta = 0.96$, 95% CI [0.80, 1.12], $p < .001$). Thus, while the provision of factual information seems to be important for attitude change, validation increased enjoyment and reduced perceptions of bias even without the provision of factual information.

Generalizability, Durability, and Behavioral Impacts

In Study 6 (n = 2,231), we tested whether the effects of sycophantic and disagreeable chatbots generalized to a non-political topic (the discussion of personal conflicts), were durable, and impacted behavior. Finally, we included a new control condition: a "neutral" chatbot prompted to neither be sycophantic nor disagreeable (see prompt table and an LLM-based text analysis of the neutral prompt in the ***Supplementary Appendix***). In other words, this control condition held the topic of discussion constant while only manipulating model behavior.

In addition to the "neutral" control condition, we kept the sycophantic, disagreeable, and undirected conditions from Studies 1-3. Participants were randomly assigned to either discuss a political opinion or an opinion about a personal conflict with the chatbot. Participants were further randomly assigned to interact with one of two popular frontier models at the time of study launch (ChatGPT 5.2 and Gemini 3; see ***Extended Data Fig. 2.*** for differences by model).

Attitude Change. As predicted, participants who interacted with the sycophantic chatbot showed approximately 2 percentage points greater attitude strength (b = 2.26, 95% CI [1.11, 3.41], $p < .001$) and 3 percentage points greater attitude certainty (b = 3.14, 95% CI [1.79, 4.50], $p < .001$) as compared to those who interacted with the neutral chatbot. By contrast, the disagreeable chatbot decreased attitude strength by approximately 6 percentage points (b = -6.44, 95% CI [-8.38, -4.50], $p < .001$) and decreased attitude certainty by 8 percentage points (b = -8.42, 95% CI [-10.58, -6.26], $p < .001$) as compared to the neutral chatbot, also supporting our pre-registered hypotheses. Thus, the findings from Studies 1-3 replicated even when using a neutral control condition and a new topic (see ***Fig. 3***).

Additional exploratory analysis reported in the ***Supplementary Appendix*** found that all key attitude change contrasts were significant when only analyzing personal topics, and three out of four were significant when only analyzing political topics. Thus, this study, combined with Studies 1-3, broadly suggests that results generalize to both personal and political topics.

Enduring Effects. To test the durability of effects, a subset of participants (n = 1,780) completed a one-week follow-up study in which we showed participants the opinions they wrote previously and had them re-rate their attitude strength and certainty. The effects of the disagreeable versus neutral chatbot persisted for both attitude strength (b = -5.68, 95% CI [-8.01, -3.34], $p < .001$) and certainty (b = -6.09, 95% CI [-8.36, -3.82], $p < .001$) at the one-week follow-up. The effects of the sycophantic versus neutral chatbot persisted for certainty (b = 2.17, 95% CI [0.33, 4.00], p = .020), but not attitude strength (b = 1.09, 95% CI [-0.68, 2.86], p = .227). Thus, three of the four key contrasts remained significant for at least a week.

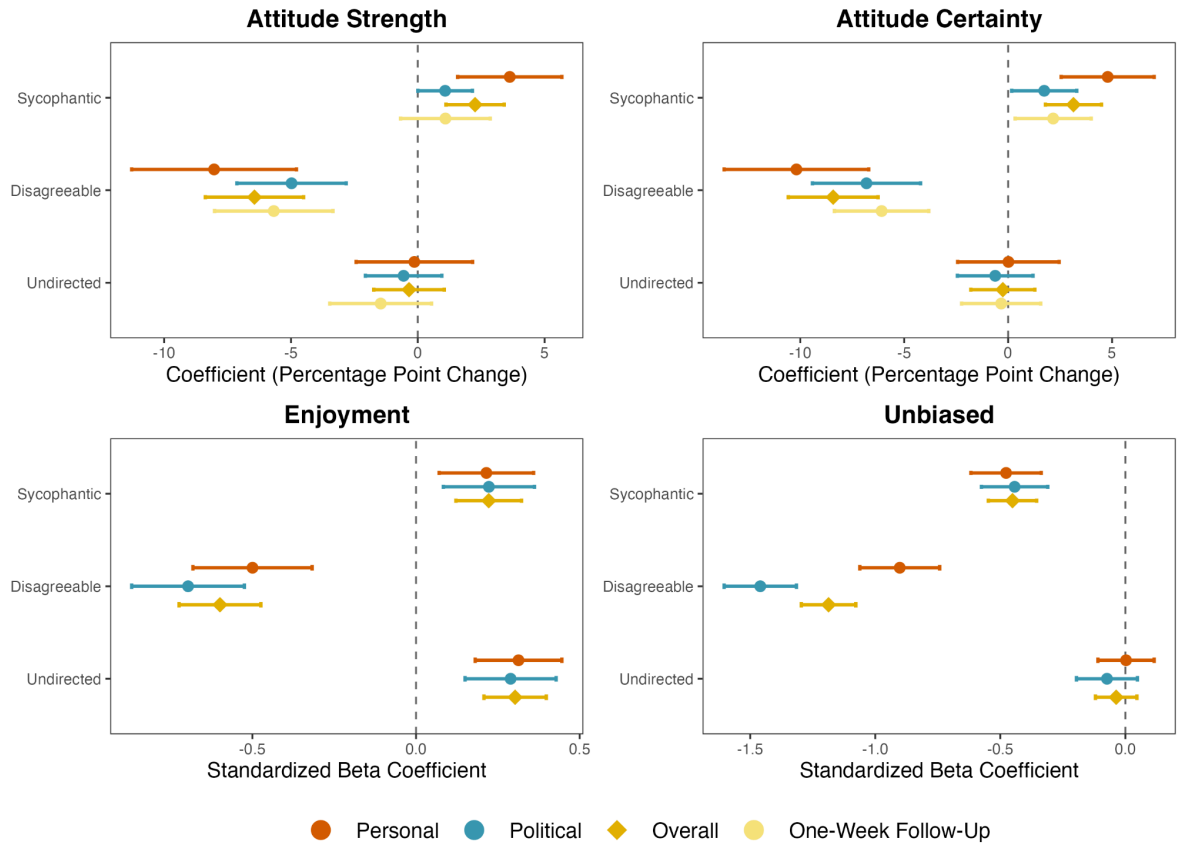


Fig. 3: Generalizability and durability of effects across personal and political topics. *The effects of AI sycophancy on our main outcome variables (attitude strength, attitude certainty, enjoyment, perceptions of bias) replicated even when using multiple topics (personal conflicts and political issues), and when using a new "neutral" control condition. The "neutral" control condition is the baseline comparison in all regressions. Additionally, when measured at a one-week follow-up, most of the effects of sycophantic and disagreeable chatbots persisted. Error bars represent 95% confidence intervals.*

Enjoyment. Also as predicted, participants enjoyed the sycophantic chatbot more than the disagreeable chatbot (β = 0.82, 95% CI [0.70, 0.94], $p < .001$) and viewed it as less biased than the disagreeable chatbot (β = 0.74, 95% CI [0.62, 0.85], $p < .001$). We also found that people enjoyed the sycophantic chatbot more than the neutral chatbot (β = 0.22, 95% CI [0.12, 0.32], $p < .001$), but there was no significant difference between the sycophantic and undirected chatbots (β = -0.08, 95% CI [-0.17, 0.01], $p = .081$). This suggests that, while people do not enjoy neutral chatbots as much as sycophantic ones, people still enjoy common AI models that are not prompted to be sycophantic.

Perception of Bias. As predicted, participants viewed the sycophantic chatbot as less biased than the disagreeable chatbot (β = 0.74, 95% CI [0.62, 0.85], $p < .001$). However, unlike Studies 1-3,

people perceived the undirected (β = -0.41, 95% CI [-0.51, -0.32], $p < .001$) and neutral (β = -0.45, 95% CI [-0.55, -0.36], $p < .001$) chatbots to be less biased than the sycophantic AI.

Behavioral Effects. To test whether the "better-than-average" effect found in Studies 1-3 translated to behavior, we devised a behavioral outcome variable whereby participants could bet small amounts of money (up to 25 cents per question) if they thought they scored better than the average participant on three tasks purported to measure desirable traits[37,38]. We administered tasks that participants were told measured empathy (the Reading the Mind in the Eyes Test)[39], intelligent and reflective thinking (the cognitive reflection test)[40], and open-mindedness (an open-mindedness scale)[41]. We then asked participants to bet money based on how they think they performed on these tasks. If participants scored better than the average participant in the study on each task, any amount of money they bet would be doubled and they would keep any leftover money. If participants scored worse than the average participant, they would lose everything they bet.

Supporting our pre-registered prediction, we found that participants bet approximately 6 cents more (out of 75 possible cents) that they were better than the average participant in the sycophantic condition as opposed to the disagreeable condition ($b = 6.23$, 95% CI [3.20, 9.27], $p < .001$), suggesting that AI sycophancy can impact costly financial decisions. When further exploring differences by condition, we found that the sycophantic AI increased the number of cents bet relative to the neutral chatbot ($b = 4.38$, 95% CI [1.38, 7.37], $p = .004$), but the disagreeable chatbot did not significantly decrease the number of cents bet ($p = .239$). Contrasts between the sycophantic and disagreeable conditions were significant for all three better-than-average betting tasks ($ps < .007$, see ***Supplementary Appendix***).

Supplementary analyses (see ***Supplementary Appendix***) found that sycophantic AI increased the number of cents bet (as compared to the disagreeable AI condition) both for those who scored below average (for two tasks: the open-mindedness measure and the cognitive reflection task, $ps < 0.013$) and for those who scored above average (for three tasks, $ps < 0.012$). Thus, for this behavioral task, sycophantic AI led to overconfident decisions for some, and justifiably confident decisions for others.

Better-Than-Average Self-Report. In contrast to the better-than-average behavioral variable, the better-than-average self-report index did not significantly differ in Study 6 when comparing the sycophantic condition to the disagreeable condition ($p = 0.142$). However, when pooling data across Studies 1-3 and Studies 6-7, there was a small but significant increase in better-than-average beliefs when comparing the sycophantic and disagreeable conditions (β = 0.16, 95% CI [0.09, 0.23], $p < .001$), as shown in ***Fig. 4.*** See ***Extended Data Figs. 3-6*** for additional outcome variables.

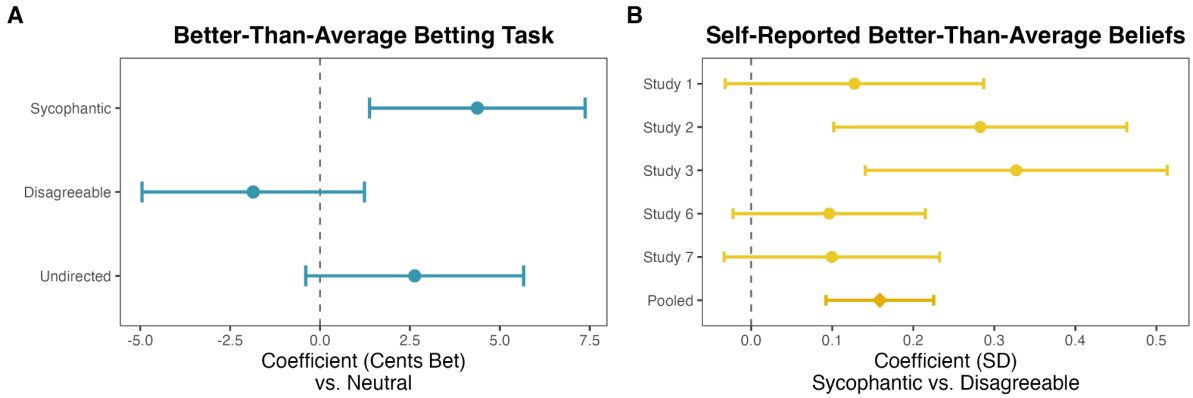


Fig. 4. Sycophantic chatbots amplified the "better-than-average" effect and impacted costly decisions. *In Study 6, participants bet more money that they were better than the average participant on tasks purporting to measure desirable traits (such as empathy and intelligence) in the sycophantic as opposed to disagreeable condition (Panel A). Additionally, when pooling data across Studies 1-3 and 6-7, participants reported slightly higher "better-than-average beliefs" in the sycophantic-versus-disagreeable AI contrast, though this effect was small (Standardized Beta Coefficient = 0.16) and not significant in every study (Panel B). We plot better-than-average beliefs only in the sycophantic versus disagreeable condition because control conditions varied across studies and because this was our pre-registered contrast of interest. Error bars represent 95% confidence intervals.*

Study 7: Testing a Chatbot that is Better at Disagreeing

Given the potential harms of chatbots that expose people to one-sided information, we tested whether we could create a "disagreeable" chatbot that provided opposing facts that people would enjoy in Study 7 (n = 1,290). Building on the findings from Study 3 that validation was associated with enjoyment while the provision of factual statements was associated with attitude change, we tested whether a "disagreeable" chatbot that presented opposing facts in a validating way would be more enjoyable. Specifically, participants were randomly assigned to three conditions: the same sycophantic and disagreeable conditions used in Study 6 (instantiated using ChatGPT 5.3), as well as a "gently disagreeable" condition in which a chatbot was prompted to simultaneously provide opposing facts and validate people's beliefs (see ***Supplementary Appendix*** for prompts and a validation of the manipulations using LLM-based text analysis).

Enjoyment. As predicted, people enjoyed the gently disagreeable chatbot more than the disagreeable chatbot ($\beta = 0.48$, 95% CI [0.34, 0.61], $p < .001$). However, people still did not enjoy the gently disagreeable chatbot as much as they enjoyed the sycophantic chatbot ($\beta = 0.16$, 95% CI [0.04, 0.27], $p = .007$). Thus, it appears that one way to make chatbots that present opposing facts more enjoyable is to accompany those opposing facts with validation.

Perceptions of Bias. Consistent with our hypotheses, people rated the gently disagreeable chatbot as less biased than the disagreeable chatbot (β = 0.57, 95% CI [0.44, 0.70], $p < .001$). However, people viewed the gently disagreeable and sycophantic chatbots as similarly unbiased ($p = .862$). In other words, people viewed chatbots that presented opposing facts as less biased if those chatbots also validated them.

Attitude Change. Participants who interacted with the gently disagreeable chatbot reported greater attitude strength ($b = 5.56$, 95% CI [3.40, 7.72], $p < .001$) and certainty ($b = 6.34$, 95% CI [3.98, 8.69], $p < .001$) than those who interacted with the purely disagreeable chatbot. However, participants who interacted with the sycophantic chatbot also reported greater attitude strength ($b = 4.51$, 95% CI [3.09, 5.93], $p < .001$) and certainty ($b = 5.23$, 95% CI [3.59, 6.87], $p < .001$) than participants who interacted with the gently disagreeable chatbot. Thus, presenting opposing facts in a gentle, validating way led to less attitude change than presenting opposing facts without validation, even if people enjoy the validating presentation more (see ***Fig. 5***):

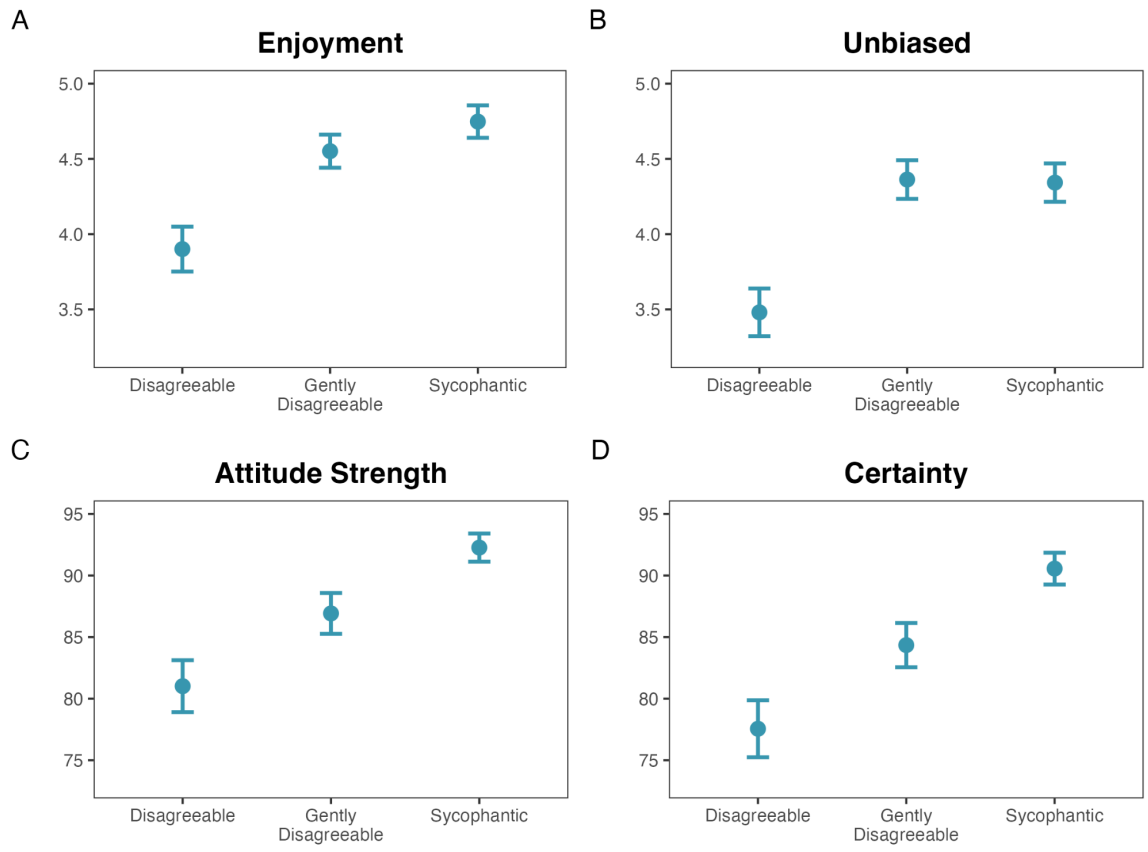


Fig. 5. Effects of a "gently disagreeable" chatbot. *Participants enjoyed the "gently disagreeable" chatbot (i.e., a chatbot prompted to present opposing information alongside emotional validation) more than the purely disagreeable chatbot, and also viewed it as less biased. However, the gently disagreeable chatbot led to less attitude change than the purely disagreeable chatbot. Error bars represent 95% confidence intervals.*

Discussion

Across seven studies, we found that interacting with sycophantic chatbots about a variety of political or personal topics led to more extreme and certain attitudes, whereas interacting with disagreeable chatbots that challenged people's beliefs led to less extreme attitudes. However, popular models that were not prompted to be sycophantic did not consistently change people's attitudes. Moreover, people enjoyed sycophantic chatbots more than disagreeable or neutral models and were more likely to choose to use them again.

People also perceived sycophantic chatbots to be more objective than disagreeable chatbots that challenged their beliefs. In contrast, third-party annotators viewed sycophantic and disagreeable chatbot replies to be equally biased. These findings align with prior literature on naive realism[42] and the "bias blind spot"[43], and extend those literatures to the domain of AI. Because people often assume that their views are objective and uncritically accept evidence that confirms their beliefs[44], they may be bad at recognizing the biases present in AI systems that agree with them. In other words, people appear to have a "bias blind spot" in identifying sycophantic AI: they are better at identifying when AI is being biased in favor of others, but not themselves. Similarly, supplementary analysis (see ***Supplementary Appendix***) found that people viewed sycophantic chatbots as both warmer and more competent than neutral or disagreeable chatbots. Efforts to ameliorate sycophancy may be particularly challenging because people rate sycophantic AI positively across a range of desirable dimensions and fail to identify its biases.

Interacting with sycophantic chatbots also led to overconfident beliefs and decisions. Specifically, sycophantic chatbots inflated people's perceptions that they were "better than average"[24] on desirable traits, such as intelligence and empathy. Furthermore, sycophantic chatbots (as compared to disagreeable chatbots) led people to bet approximately 6 cents more (out of 75 total cents) that they were better than the average participant on tasks purporting to measure desirable traits, demonstrating that sycophantic AI can impact costly decisions. Supplementary Analysis (see ***Supplementary Appendix***) found that sycophantic AI also led to overconfident perceptions of knowledge: specifically, participants who interacted with sycophantic chatbots rated themselves as more knowledgeable on the topic they discussed with the AI, even though sycophantic chatbots had no effect on an objective knowledge quiz.

When breaking up different components of sycophancy, we found that sycophantic and disagreeable AI led to more attitude change when they were prompted to include factual statements. This aligns with recent work suggesting that the provision of factual information is a key factor driving AI's persuasive ability[9,30]. While chatbots that provided validation (without factual statements) did not significantly change attitudes, they were rated as more enjoyable and less biased. Building on these findings, a final study found that accompanying opposing facts

with emotional validation increased enjoyment and reduced perceptions of bias. Thus, one promising way to design AI systems that encourage openness to opposing perspectives while keeping interactions enjoyable is to present those perspectives in a validating manner.

Despite research suggesting that chatbots can frequently be sycophantic[4], the undirected models (or the models with no additional prompting) did not lead to increased attitude extremity and overconfidence in short-term interactions. Thus, the effects we describe in this paper mainly pertain to strongly sycophantic chatbots as opposed to default model behavior. However, supplementary analyses (see ***Supplementary Appendix***) of a separate dataset of naturalistic ChatGPT interactions found that ChatGPT replies can sometimes be as sycophantic (or validating) as the replies tested in our sycophancy condition. For instance, among conversations that lasted at least 25 turns, 46% contained at least one chatbot reply that was approximately as validating as the typical reply in the sycophancy condition. Other work has documented high levels of sycophancy when users discuss certain topics[35] or in highly personalized interactions[5,36]. Altogether, this work suggests that, even though sycophancy is not default model behavior and will vary across contexts, models, or time[35], it occurs frequently enough to take its risks seriously. This paper illustrates the causal effects of exposure to strongly sycophantic AI behavior.

While some studies suggest that AI can expose people to more accurate[45] or politically moderate[46] viewpoints, our work suggests that people often prefer to use AI that confirms their pre-existing views[15], which can in turn make people's beliefs more extreme.
Indeed, the very qualities that help AI be effective at debunking falsehoods, such as its ability to provide targeted facts and evidence[30], may make it an effective tool for creating rationalizations of one's own pre-existing beliefs[15]. Consistent with theories of motivated cognition[47] and confirmation bias[48], whereby people seek out and interpret information in a biased way, our work finds that people prefer belief-confirming chatbots and dismiss information from counterattitudinal chatbots as biased[15]. This research reveals the psychological and behavioral impacts of AI sycophancy while also revealing how to create non-sycophantic chatbots that people enjoy using.

Methods

Data Availability and Ethics. Pre-registrations, code, materials, and data are all available on the Open Science Framework (OSF): https://osf.io/5k6r9. All experiments (Studies 1-3 and 6-7) were pre-registered. All studies received IRB approval via the NYU Ethics Review Board (IRB-FY2025-10299). Raw conversation data are omitted for privacy-preserving reasons.

Study 1. A politically-balanced (half liberal, half conservative) and gender-balanced sample of 1,081 US participants (Male = 537, Female = 534, Non-binary/Other = 10, M_{age} = 38.34, SD = 13.19, Liberal = 509, Conservative = 572) was recruited from the survey platform Prolific Academic. Recruitment began on June 19, 2025. Following our pre-registered exclusion criteria, we automatically screened out participants who did not consent, failed a captcha, and did not pass a simple attention check at the beginning of the survey (i.e., a simple question asking "You have two apples, three grapes, and an orange. How many pieces of fruit do you have?"). We also screened out an additional 24 participants who failed an attention check administered pre-treatment. We based our sample size on an a priori power analysis using G*Power that suggested we would need a sample size of 253 participants per cell to detect a small effect of d = 0.25 with 80% power (see pre-registrations for power analysis: https://osf.io/5k6r9).

Procedure. Participants participated in a short conversation about a polarized political issue (gun control) with an AI chatbot powered by "gpt-4o" (called via the OpenAI API) that was embedded directly within the Qualtrics survey platform using a tool called "Vegapunk"[10]. Participants first indicated 1) how much they supported gun control on a scale from 0 (strongly disagree) to 100 (strongly agree), and 2) how certain they were about their opinions about gun control on a scale from 0 (very uncertain) to 100 (very certain).

Then, participants were randomly assigned to four experimental conditions. In one condition, participants interacted with undirected gpt-4o. In another condition, participants interacted with a sycophantic AI chatbot that we prompted to validate and affirm their perspective. In a third condition, they interacted with a disagreeable AI chatbot that we prompted to question their beliefs and open them up to alternate perspectives. In a control condition, participants talked to an AI chatbot about the benefits of owning dogs and cats, following prior work[8]. The prompts provided to ChatGPT-4o for the chatbot conversations are shown in the ***Supplementary Appendix***.

Afterwards, participants answered the same questions about their support for gun control and belief certainty. Then, they were asked a number of other post-treatment questions. In particular, participants answered two questions about their enjoyment of their AI conversation and desire to interact with the AI chatbot in the future on a Likert scale from 1 (strongly disagree) to 6 (strongly agree). These two questions were averaged into a single summary score of AI

enjoyment. Participants also rated the extent to which the AI was unbiased on the same Likert scale. See the ***Supplementary Appendix*** for phrasing of all key outcome variables. All experiments (e.g. Studies 1-3 and Study 6) were pre-registered.

Analysis. Following our pre-registration and past work[8], we conducted linear regression models with heteroskedasticity-robust standard errors testing the impact of the treatment condition on the post-treatment outcomes, controlling for pre-treatment measures when applicable. The control condition was the reference group in all models. We followed up these regression models with several pre-registered planned contrasts using t-tests on the estimated marginal means from the main model. While we were interested in comparisons to the control condition, we were also interested in contrasts between the sycophantic and disagreeable conditions (particularly for detecting smaller effects), and in comparisons between the sycophantic/disagreeable condition and the undirected AI (to help understand baseline AI models without additional prompting).

For the 100-point scales (extremity and certainty) we report unstandardized coefficients, but we report standardized beta coefficients (β) for other variables (bias and enjoyment). For the extremity variable, we defined attitude extremity as distance from the midpoint (50) on the 0-100 support for gun control scale, as pre-registered. We considered this to be the most theoretically meaningful variable, since we were interested in whether sycophantic AI would make people's beliefs more extreme (far from the midpoint) as opposed to moderate (close to the midpoint). Nevertheless, we report changes in the overall belief distribution, as well as plots showing distributions of beliefs, in the ***Supplementary Appendix***. We repeated this same analysis for studies 2 and 3.

Study 2. A representative quota sample (by age, sex, and political party) of 1019 US participants (Male = 500, Female = 509, Non-binary/Other = 10, M_{age} = 45.64, SD = 15.79, Liberal = 555, Conservative = 464) was recruited via Prolific Academic. Recruitment began on August 11, 2025. Once again following our pre-registered exclusion criteria, we automatically screened out participants who did not consent and failed a bot check or CAPTCHA. Also following our pre-registration, we screened out an additional 24 participants who failed an attention check and an additional 40 participants who reported technical errors with the chatbot.

The procedure was identical to *Study 1*, but this time, participants were randomly assigned to three different political issues (abortion, immigration, and healthcare) and we used a newer model ("gpt-5-chat" called via OpenRouter) to examine whether the results would generalize. To test if our self-reported results translated to meaningful behavior, we included an additional behavioral task in which we gave participants the option to speak with the same chatbot again about a new topic. Finally, we gave participants a 10-item knowledge quiz at the end to see if the different chatbots impacted factual knowledge. The prompts were also slightly updated so that

we did not ask the AI to produce responses with roughly the same length each time (see ***Supplementary Appendix*** for a table of all prompts used).

Study 3. A semi-representative quota sample (by age, sex, and political party) of 1183 US participants (Male = 563; Female = 609, Non-binary/Other = 11; M_{age} = 44.53, SD = 15.00, Liberal = 602, Conservative = 581) was recruited via Prolific Academic. Following our pre-registration, we loosened the representative quota after 48 hours of recruiting (for the last 132 places in the study) to speed up recruitment. Also following our pre-registration, we excluded an additional 34 participants who failed an attention check and an additional 107 participants who reported technical issues with the chatbot.

Participants once again discussed the topic of gun control, as they did in *Study 1*. They were randomly assigned to five experimental conditions: 1) a sycophancy + facts condition in which the AI chatbot was prompted to validate the user and provide them with facts to support their perspective; 2) a sycophancy + no facts condition in which the AI chatbot was prompted to validate the user but not discuss any facts; 3) a disagreeable + facts condition, in which the AI chatbot was prompted to question the user and provide facts to support that questioning; 4) a disagreeable + no facts condition, in which the AI chatbot was prompted to question the user but not discuss any facts; and 5) a control condition in which the AI chatbot discussed the pros and cons of owning cats versus dogs. This experiment also randomly assigned people to interact with three different AI models ("claude-sonnet-4", "gemini-2.5-flash," "gpt-5-chat" called via their APIs using OpenRouter) to test generalizability.

Pooled Analysis. Because results were consistent across studies, we present the pooled results of Studies 1-3 in the main text. See the ***Supplementary Appendix*** for prompt table, all outcome variables, the results for individual studies, descriptive statistics, robustness checks employing Bonferroni corrections, differential attrition analysis, and robustness analyses using beta regressions. All individual studies were pre-registered, but the pooled analysis was not. The pooled analyses excluded the "non-fact-providing" sycophantic and disagreeable conditions from Study 3 because factual information appeared to play an important role in attitude change, and the provision of factual information was artificially removed in these conditions (see the ***Supplementary Appendix*** for descriptive statistics on the amount of factual information in the fact-providing and non-fact-providing conditions, as rated by human annotators). However, an additional robustness check shown in the ***Supplementary Appendix*** found that key conclusions remained unchanged when including the non-fact-providing chatbots in the pooled analysis.

Study 4. A representative quota sample (by age, sex, and political party) of 403 US participants (Male = 200; Female = 198; Non-binary/Other = 5; M_{age} = 45.66, Liberal = 217, Conservative = 186) was recruited via Prolific Academic. Recruitment began February 10, 2026. These participants rated 504 randomly selected AI chatbot replies (with an equal number of replies

from each condition) from Studies 1-3. Each chatbot reply was rated by 8.00 (*SD* = 2.97) participants on average. The previous message was provided before each chatbot reply for context. Participants rated the extent to which they viewed conversations as 1) containing social validation, 2) containing factual information, and 3) being unbiased. Participants also rated different dimensions of sycophancy (e.g., the amount of flattery or agreement in a reply) and the extent to which humans asked AI to respond differently (e.g., to be more or less sycophantic). See the ***Supplementary Appendix*** for questions supplied to participants and additional analysis of Study 4 data.

Analyses. To analyze differences by condition in human annotator perceptions, we ran regression analyses exploring the key variables as a function of the experimental condition, while controlling for study, topic, and model (mirroring the models in Studies 1-3). Standardized regression results were reported for ease of interpretation.

Study 5. A sample of 20 US workers pre-screened by Prolific as being qualified for rating the factual accuracy of AI output were recruited (M = 11, F = 8, Non-binary/Other = 1, M_{age} = 40.85, SD_{age} = 14.42, Liberal = 16, Conservative = 4). These participants fact-checked a total of 191 replies from Studies 1-3 (balanced across experimental condition). Messages were seen by 2.61 (*SD* = 1.25) fact-checkers on average.

Study 6. A semi-representative quota sample of 2,231 US participants (M = 966, F = 1246, Non-binary/Other = 19, M_{age} = 40.43, SD_{age} = 13.94, Liberal = 1,268, Conservative = 963) was recruited via Prolific Academic. Recruitment began on March 1, 2026. Following our pre-registration, we aimed to recruit 2,500 participants. 2,502 participants filled out our survey, though because we unintentionally made our survey briefly available to the same participants multiple times, we had to exclude 11 duplicate Prolific IDs, leaving us with a total of 2,491 participants. Following our pre-registration, we also loosened our demographic quotas after 48 hours (and after collecting 1,100 participants) to recruit a more general US sample. Following our pre-registered exclusion criteria, 37 participants who failed our bot check were automatically screened out, another 99 were excluded who failed an early pre-treatment attention check, and another 124 participants who reported technical issues with the chatbot were excluded, leaving us with a total sample size of 2,231 participants.

Participants were randomly assigned to discuss a 1) political topic or 2) a recent and serious personal conflict. Participants could choose from a list of political topics or personal conflicts they wanted to discuss, or provide their own. Of those randomly assigned to discuss a political topic, 29.2% picked abortion, 29.1% picked healthcare, 19.2% picked immigration, 17.8% picked gun control, and 4.6% picked “Other.” Of those assigned to discuss a personal topic, 30.3% chose to discuss a conflict with a romantic partner, 29.7% chose to discuss a conflict with a family member, 19.1% chose to discuss a conflict with a friend, 16% chose to discuss a conflict

with a work colleague, and 4.9% picked "Other." See the ***Supplementary Appendix*** for an analysis of differential effects by topic. Participants then indicated how 1) strongly they held their opinion on a scale of 0-100, and 2) how certain they were in the opinion on a scale of 0-100. Because participants were choosing their own political and personal issues (rather than being assigned a specific issue to discuss), we could not define "attitude extremity" as distance from the midpoint, since the midpoint would differ for each person. Instead, we measured attitude strength, which looked at how strongly participants held the particular opinion they chose to discuss. We note that attitude extremity is considered to be a type of attitude strength[49], meaning that we are looking at similar constructs across studies.

Then, participants were randomly assigned to four experimental conditions. As in Studies 1-3, we retained the same sycophantic, disagreeable, and undirected chatbot conditions. The sycophantic "fact-providing" prompts were used for these studies. However, we also added a new "neutral" chatbot condition in which participants interacted with a chatbot that was prompted to neither validate nor disagree. See ***Supplementary Appendix*** for the full prompt and a validation of this neutral control condition. Participants were further randomly assigned to interact with two different models: "gpt-5.2-chat" or "gemini-3-flash-preview" (as called via the OpenRouter API). In contrast to Studies 1-3, in which participants were constrained to only three turns, participants could choose to end the conversation after two conversational turns or continue for up to 8 conversational turns.

After the chatbot conversation, participants once again indicated their strength and certainty about their opinion about the political topic or personal conflict. Participants answered the same questions about enjoyment, bias, warmth, competence, and the "better-than-average" effect administered in Studies 1-3. Further, participants completed a behavioral outcome variable assessing the better-than-average effect. Specifically, participants completed two items from the reading the mind in the eyes task, two items from the cognitive reflection task, and a scale measuring open-minded thinking. Then, participants were asked to bet up to 25 cents (for each of the three questions) that they scored better than the average participant on these tasks. If participants did score better than the average participant, any amount of money bet would be doubled. If participants did not score better than the average participant, participants would lose all money bet. Participants kept any money they did not bet.

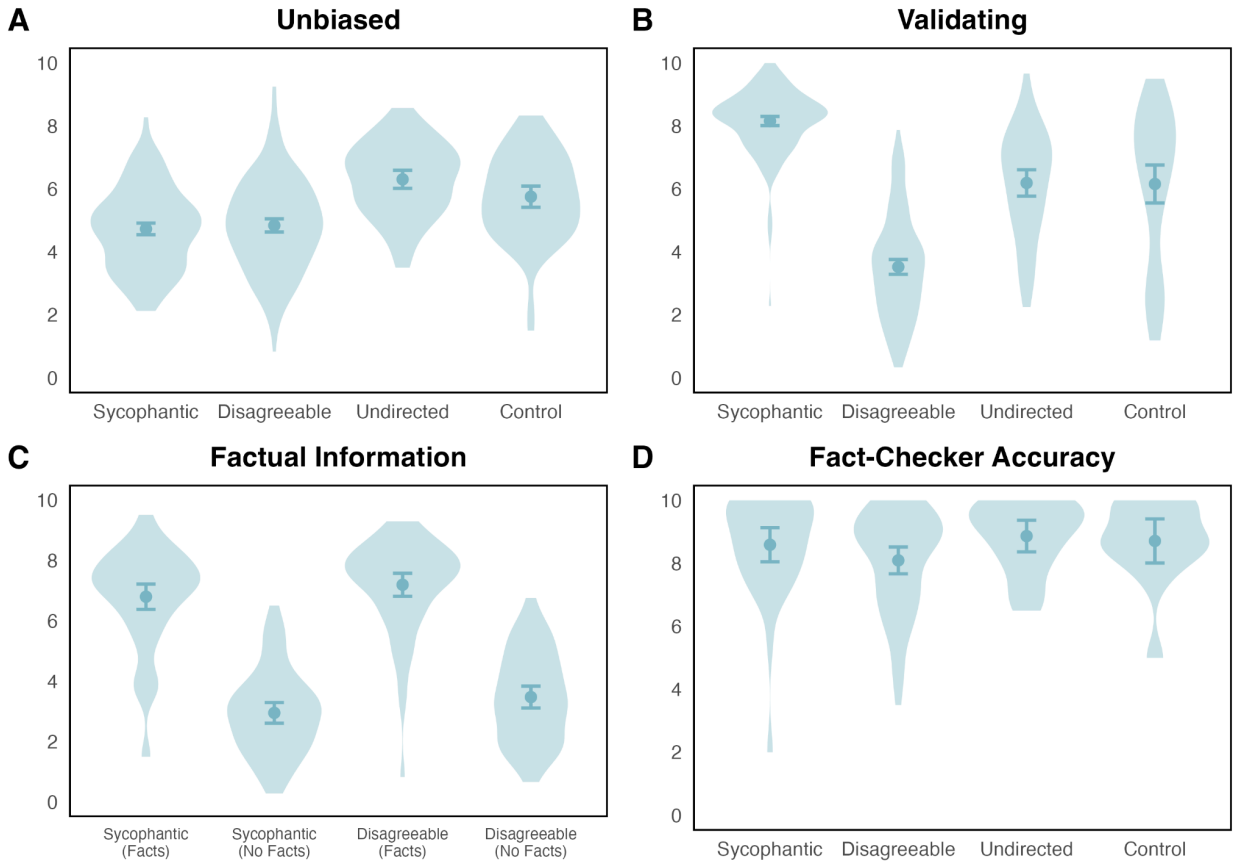
Study 6 Follow-Up Survey. A one-week follow-up study was then sent out to participants on March 8, 2026. Data collection stopped one week later, on March 16, 2026. 1,780 participants (M = 784, F = 977, Other = 19; M_{age} = 41.09, SD_{age} = 13.91, Liberal = 1003, Conservative = 777) filled out the follow-up survey (79.8% retention). Participants were shown the opinion regarding the personal or political topic they expressed during the earlier survey and were told to rate their strength and certainty about the opinion again.

Study 7. A semi-representative quota sample (by age, sex, and political party) of 1,290 US participants (M = 665, F = 614, Other = 11; M_{age} = 41.52, SD_{age} = 15.38, Liberal = 639, Conservative = 651) was recruited via Prolific Academic. Recruitment began on April 6, 2026. Following our pre-registration, we aimed to recruit 1,500 participants and ended up recruiting 1,497 participants. As pre-registered, we loosened our representative quota after 48 hours and after collecting 1,292 participants. 23 participants were automatically screened out for failing an attention/bot check, 66 participants were further excluded for failing a pre-treatment attention check, and an additional 118 participants were then excluded for reporting technical errors with the chatbot. This left us with a total of 1290 participants.

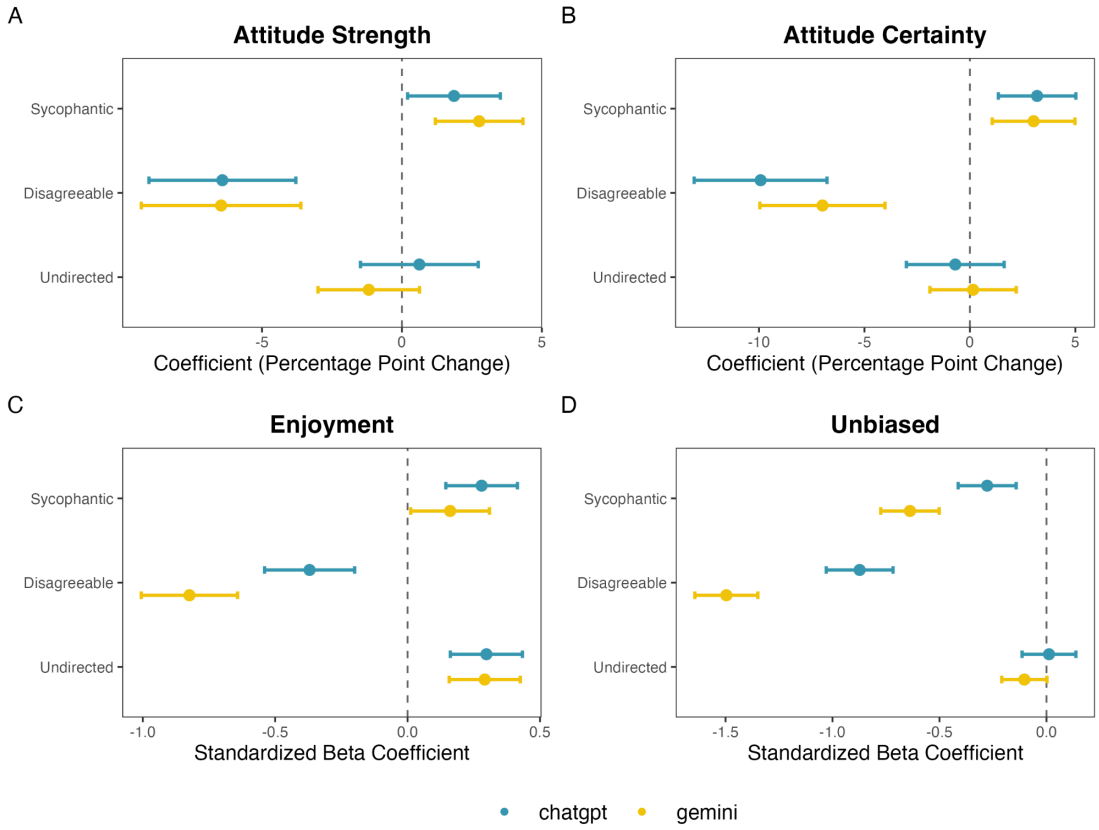
Participants were then randomly assigned (as in Study 6) to discuss a personal or political topic with an AI chatbot. They either spoke to a sycophantic or disagreeable chatbot (using the same sycophantic and disagreeable fact-providing chatbots used in other studies and powered by "chatgpt-5.3" called via OpenRouter), or to a "gently disagreeable" chatbot that would first validate a user's belief and then present opposing facts in an emotionally validating and non-threatening way (see prompt table for the prompt behind this chatbot in the ***Supplementary Appendix***). Participants could choose to end the conversation after two conversational turns or continue for up to 8 conversational turns. Participants then completed the same key outcomes measured in Study 6.

Moderation Analysis. For the moderation analysis, we pooled data from Studies 1-3 and 6. We tested for an interaction between each moderator and condition on each outcome, controlling for study, AI model, and topic, using heteroskedasticity-robust standard errors as in prior models.

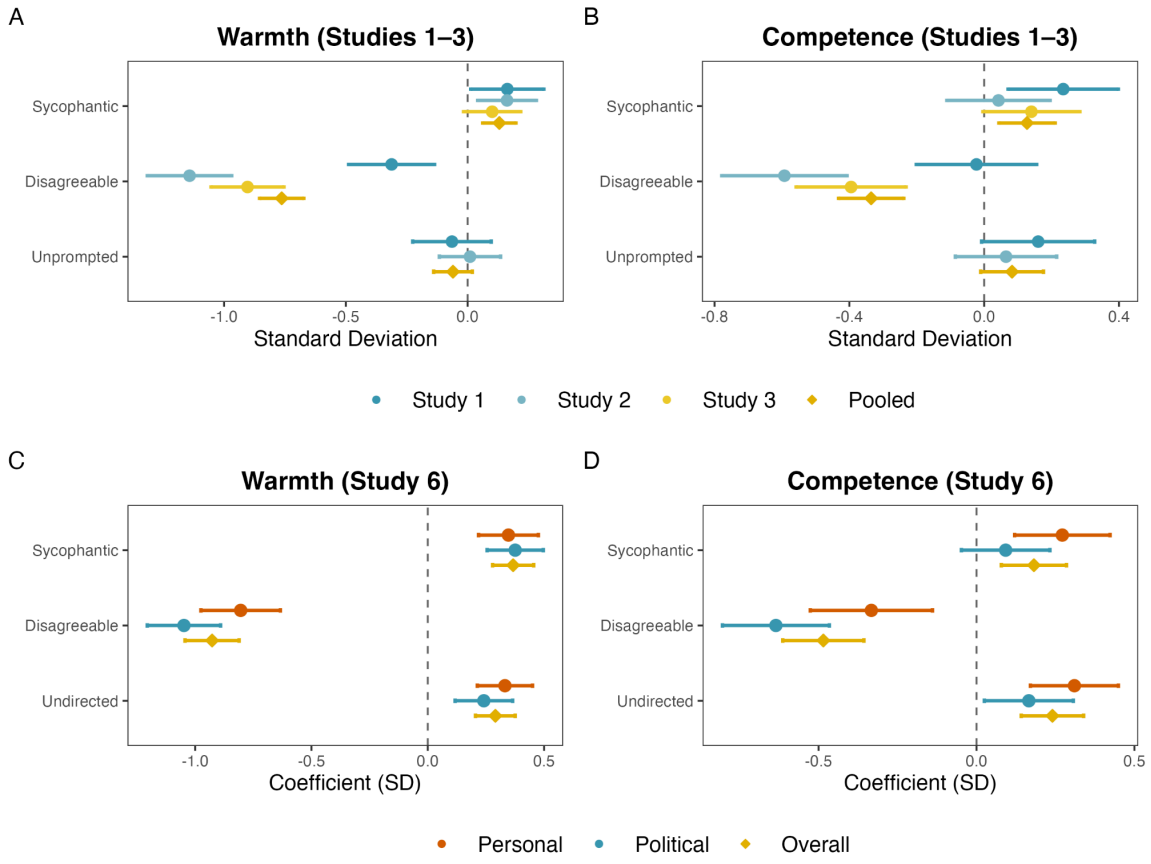
Extended Data Figures



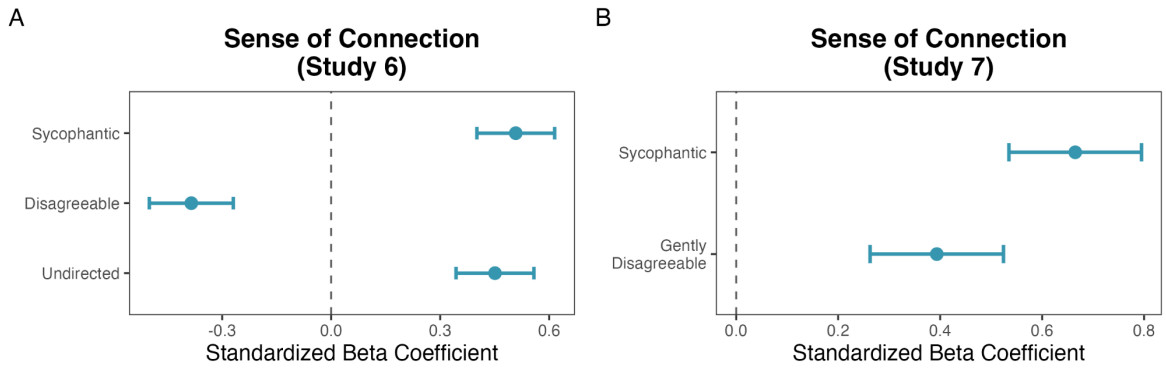
Extended Data Fig. 1. Lay annotator and fact-checker judgements of chatbot replies. While participants perceived sycophantic chatbots as less biased than disagreeable chatbots, lay third-party observers viewed these chatbots as equally biased (Panel A), suggesting a potential "bias blind spot" in perceptions of sycophantic chatbots. Lay annotators also confirmed that our prompt-based manipulations of validation (Panel B) and factual information (Panel C) were effective. Qualified fact-checkers rated the facts provided by each chatbot as mostly accurate, with no significant difference by condition (Panel D). Panel C only shows the fact-providing and non-fact-providing conditions shown in Study 3. Panels A, B, and C include both the regular sycophantic and disagreeable conditions, as well as the sycophantic/disagreeable and fact-providing/non-fact-providing conditions from Study 3. Error bars represent 95% confidence intervals, and distributions are plotted with violin plots.



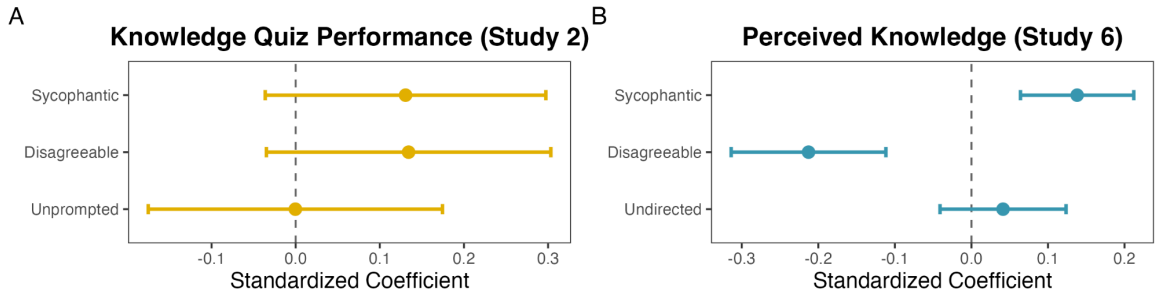
Extended Data Fig. 2. Model differences. Effects for the Study 6 outcome variables, with results broken down by model (ChatGPT vs. Gemini).



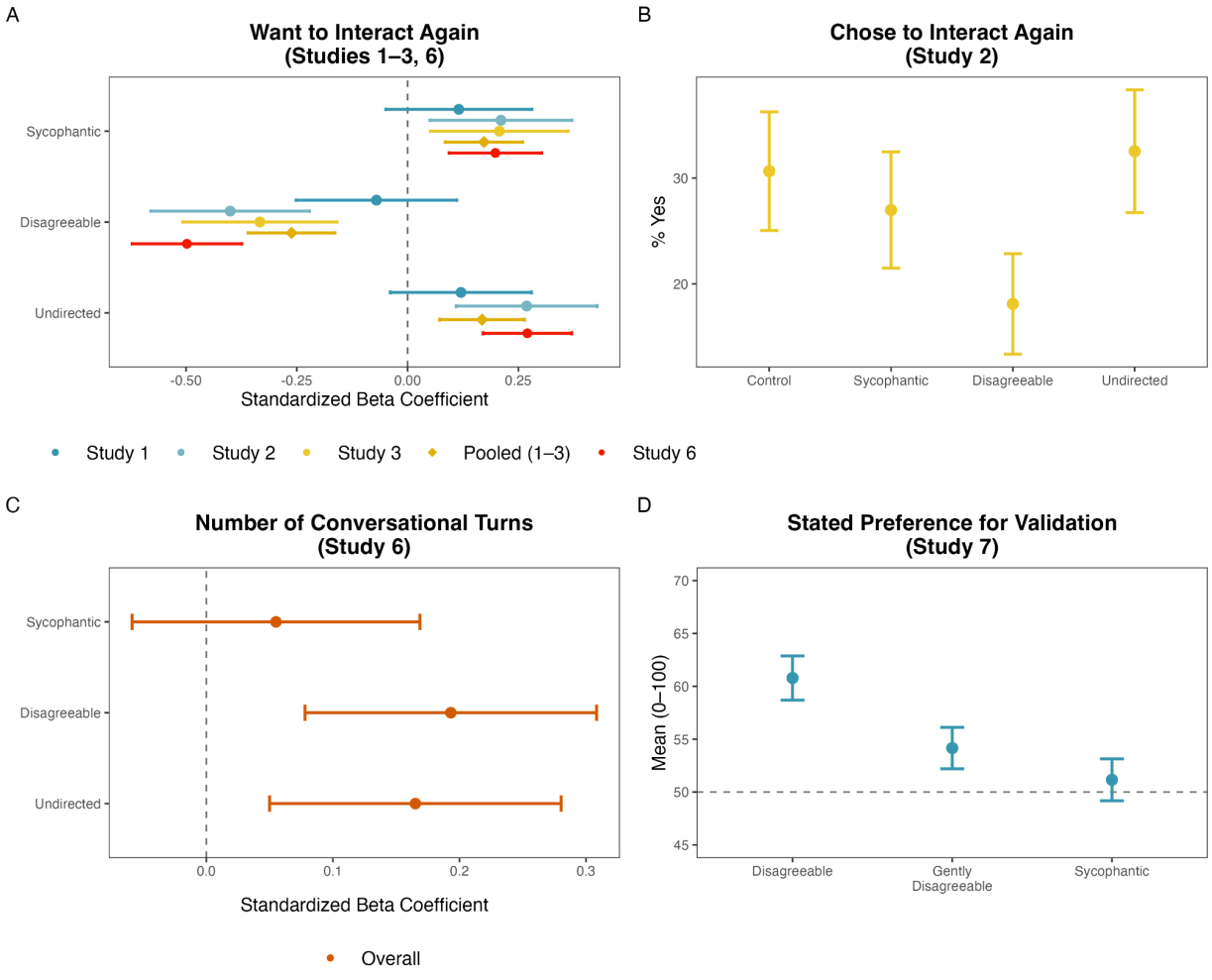
Extended Data Fig. 3. Warmth and competence. Across studies, participants viewed the sycophantic AI as warmer and more competent than the control condition used in Studies 1-3 in which people discussed an unrelated topic with an undirected AI and the disagreeable AI (Panels A and B). They also viewed the sycophantic AI as warmer and more competent than neutral AI and disagreeable AI in Study 6 (Panels C and D).



Extended Data Fig. 4. Sense of connection. People reported a stronger sense of connection with the sycophantic AI as compared to the neutral AI and disagreeable AI, suggesting that sycophancy may be a part of a broader class of relationship-seeking behaviors.



Extended Data Fig. 5. Perceived versus actual knowledge. The AI chatbot type did not influence performance on a 10-item objective knowledge quiz about the topic at hand (e.g., abortion, immigration, or healthcare in the US) (Panel A). However, sycophantic AI significantly increased how much people perceived they knew about the topic at hand as compared to neutral AI, whereas disagreeable AI significantly decreased how much people perceived they knew about the topic at hand as compared to neutral AI (Panel B). The neutral and undirected AI did not significantly differ. Thus, while sycophantic AI does not improve actual knowledge, it increases people's perceived knowledge.



Extended Data Fig. 6. Different Measures of Engagement. Participants reported wanting to interact with the sycophantic AI again as compared to the disagreeable chatbot in a self-report outcome variable in Studies 1-3 (Panel A) and Study 6 (Panel B). In a measure of behavioral engagement, participants were less likely to choose to interact with the disagreeable chatbot as compared to other chatbots (Panel C). However, when examining the number of conversational turns participants chose to take with the AI chatbots (Panel D), participants engaged in more conversational turns with the disagreeable and undirected AI (but not the sycophantic AI) as compared to the neutral AI.

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